

PRATIK SANJAY KHAIRNAR

Cybersecurity Fresher - VAPT (Vulnerability Assessment and Penetration Testing) Engineer

Authentication and Password-Reset Vulnerability Research | Security Tool Development

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PROFESSIONAL SUMMARY

Final-year B.Tech Cyber Security and Forensics student and Junior Penetration Tester with hands-on focus on authentication and password-reset vulnerability exploitation - token predictability, host-header poisoning, session fixation, and OTP/2FA-flow tampering - built through live VAPT engagements and deepened by independently designing and building AUTHENTIX v2.1.0, a Burp Suite extension (Java, Montoya API) with 27 evidence-gated security invariants and 28 secret rules that models and tests authentication workflows end-to-end, validated against a purpose-built vulnerable lab target. Also builds Python security tooling (Reflectra, a browser-confirmed XSS scanner; BlindStrike, a blind SQL injection framework) to automate vulnerability discovery. Seeking an entry-level VAPT Engineer / Penetration Tester role to apply hands-on exploitation, tooling, and reporting skills.

FLAGSHIP PROJECT - AUTHENTIX

AUTHENTIX - Authentication Security Research Platform for Burp Suite

2026

Independent Research | Java, Burp Suite Montoya API, Swing, JUnit 5 | Private repository, code and live demo available on request

- Designed and built a Burp Suite extension focused on authentication-workflow security (login, OTP, session, password reset, account recovery) - an area not directly targeted by authorization-focused tools such as AuthMatrix and Autorize.
- Built a token-lifecycle tracker (SHA-256 fingerprinted, raw secrets never stored), a per-actor authentication flow graph, and 27 evidence-gated security invariants - including OTP replay, OTP client-trust bypass, session fixation, reset-token reuse, host-header reset poisoning, JWT structural weaknesses/algorithm-confusion surfaces, refresh-token rotation, and cross-actor account-enumeration detection.
- Built an attack-path correlator that chains confirmed findings into real-world attack narratives, integrated 28 secret analysis rules, and engineered a Swing UI (Findings, Attack Chains, Flow Graphs, Token Registry) with one-click Markdown export.
- Validated detection logic with a standalone JUnit 5 harness (23/23 expected findings, zero false positives on secure traffic) and against NimbusBank v2 testbed with 16 working, exploitable vulnerabilities, each confirmed with live HTTP requests.

SECURITY TOOLING PROJECTS

Reflectra - Context-Aware XSS (Cross-Site Scripting) Scanner

v7.0.0

Python, Selenium, Headless Chrome | github.com/pratik-khairnar-sec/Reflectra

- Confirms Cross-Site Scripting (XSS) via real headless-Chrome JavaScript dialog execution, not response string-matching, layered with a context-aware HTTP reflection probe that prioritizes about 2,600 payloads by injection context without dropping coverage on ambiguous probes; adds DOM-sink analysis, authenticated and blind XSS scanning, thread-pooled scanning, and Telegram-delivered HTML reports.

BlindStrike - Blind SQL Injection Testing Framework

v7.0.0

Python | github.com/pratik-khairnar-sec/BlindStrike

- Detects time-based (SLEEP, WAITFOR, pg_sleep) and boolean-based (TRUE/FALSE content-diff, noise-floor normalized) blind SQL injection across MySQL, MSSQL, PostgreSQL, and Oracle, with baseline measurement and confirmation retest; supports URL, POST body, and header injection via a FUZZ marker, multi-threaded scanning, a guided CLI wizard, and JSON/CSV/HTML reporting with real-time Telegram alerts.

PROFESSIONAL EXPERIENCE

Junior Penetration Tester - GROOTT Services

Jun 2025 - Oct 2025

- Performed manual web application penetration testing with Burp Suite to identify, validate, and document OWASP Top 10 vulnerabilities, including broken authentication, broken access control, and injection flaws.
- Tested authentication workflows, including password reset, session management, and login, for logic flaws and token-handling weaknesses; compiled VAPT reports with CVSS-aligned risk ratings and retested fixes before client delivery.

Cyber Crime Intern - Cyber Police Station, Nashik Police Commissionerate

Jan 2026 - May 2026

- Investigated 15+ live cybercrime cases, applying digital forensics techniques for incident investigation and cyber fraud detection; maintained strict chain of custody for digital evidence.
- Conducted network traffic analysis and OSINT investigations to trace ownership behind suspicious websites; performed malware analysis on Android APK evidence and documented indicators of compromise (IOCs) for senior officers.

TECHNICAL SKILLS

Authentication and Access Control: Password Reset Vulnerabilities, Broken Authentication, Session Management, OTP/2FA Bypass, IDOR, JWT Analysis

Web Application Security: VAPT, Manual and Automated Penetration Testing, OWASP Top 10, OWASP ASVS, SQL Injection, Cross-Site Scripting (XSS), Vulnerability Assessment

Security Tool Development: Python (Selenium, Requests, Threading, argparse), Java (Burp Montoya API, Swing, JUnit 5), Telegram Bot API Integration

Security Tools: Burp Suite (including extension development), Nmap, Nessus, Metasploit, SQLMap, Ghauri, Nuclei, Subfinder

Digital Forensics and Investigation: OSINT, Malware Analysis, Network Traffic Analysis, Chain of Custody, Incident Response

Reporting and Standards: CVSS Scoring, Risk Prioritization, Remediation Guidance | Networking and OS: HTTP/HTTPS, TCP/IP, DNS, Linux, Windows

EDUCATION & CERTIFICATIONS

B.Tech, Cyber Security and Forensics - Sandip University, Nashik

2023 - Present

- Certified Ethical Hacker (CEH v13) - July 2026 | Cisco Certified Network Associate (CCNA) - November 2025